



Proxmox: Storage, Backup, Capacity, and Naming

Purpose: Documents the virtualisation host's storage layout, the backup and recovery strategy, the capacity plan for the planned estate, and the lab-wide machine naming convention. Written so that a reviewer can see not just *what* was configured but *why*, and which controls it satisfies.

Host: proxmox-01 (192.168.10.6), Proxmox VE 9.2.2, standalone node.

Status: Backup configured and verified 2026-08-30.

1. Host overview

Resource	Value
CPU	8 cores/threads
RAM	~32 GiB
Storage	2 physical disks, 2.67 TiB total
Cluster	Standalone (no cluster)

RAM is the binding constraint for how many guests run at once; CPU and storage are abundant.

The screenshot displays the Proxmox VE 9.2.2 web interface. The left sidebar shows the 'Datacenter' view with a tree structure including 'proxmox-01' and its associated storage and network resources. The main panel shows the 'Health' status with a green checkmark and 'Standalone node - no cluster defined'. Below this, the 'Nodes' section shows 1 online node and 0 offline nodes. The 'Guests' section shows 0 running and 2 stopped virtual machines, and 0 running and 0 stopped LXC containers. The 'Resources' section shows CPU, Memory, and Storage usage. At the bottom, a 'Tasks' table shows a recent task: 'Update package database' completed successfully at 06:44 on Aug 29.

Start Time	End Time	Node	User name	Description	Status
Aug 29 06:44	Aug 29 06:48	proxmox-01	root@pam	Update package database	OK

Figure 14.1: Datacenter summary: a healthy standalone node (`proxmox-01`), no cluster, with the current guests. This is the top-level health and inventory view.

2. Storage layout

Disk	Size	Role
<code>/dev/sda</code>	3 TB	Proxmox OS + primary guest storage (<code>local</code> , <code>local-lvm</code>)
<code>/dev/sdb</code>	500 GB	Backup target (<code>backup</code> , ext4 Directory storage)

`sda` holds the two default install storages: `local` (ISO images, templates, backups directory) and `local-lvm` (VM/CT disk images). `sdb` was previously an OS disk (leftover EFI + ext4 partitions); it was wiped and repurposed as a dedicated backup store.

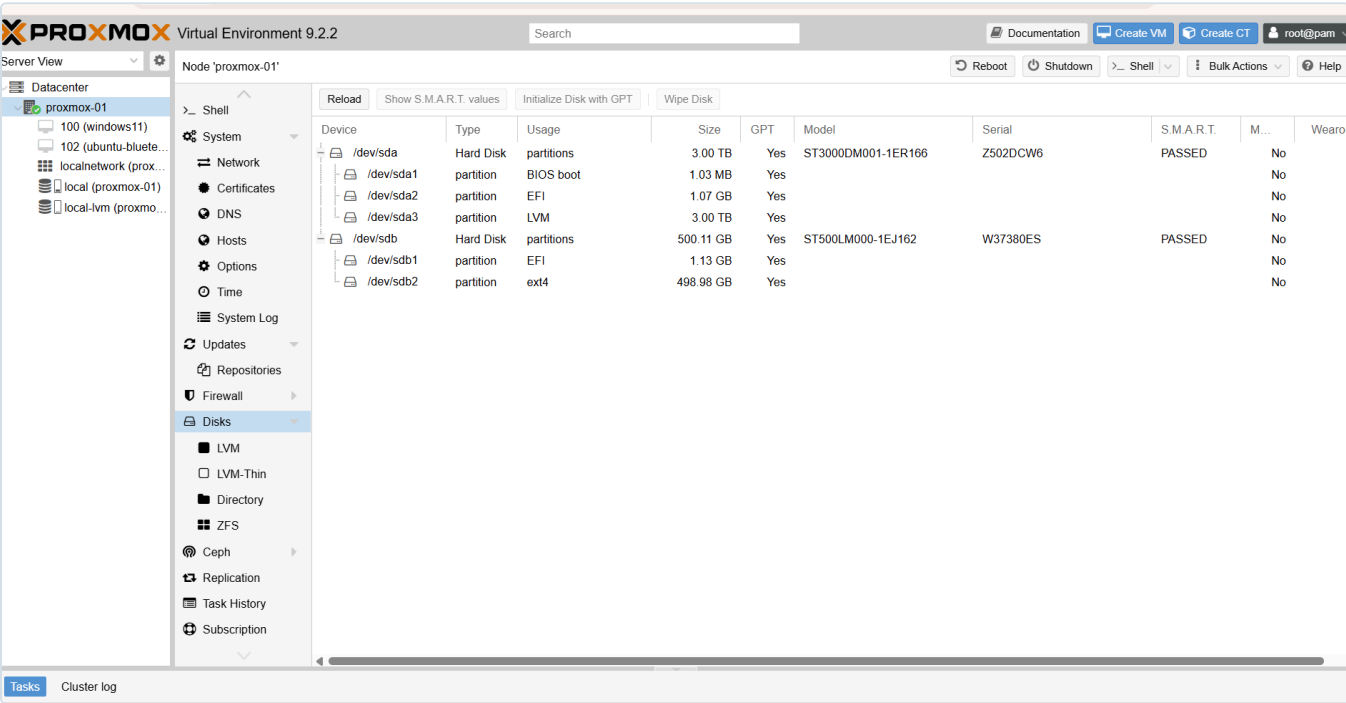


Figure 14.2: The two physical disks: `/dev/sda` (3 TB, Proxmox OS + LVM guest storage) and `/dev/sdb` (500 GB), before `sdb` was repurposed. Both report SMART "PASSED".

Why backups go on a separate disk

Keeping backups on a *different physical disk* from the running VMs means a failure or corruption of the primary disk does not take the backups with it. This is a basic tenet of recoverable design: the backup must survive the thing it is protecting against.

3. Backup and recovery strategy

The principle

A running VM is not a backup of itself. Snapshots help with quick rollback, but a real backup is an independent, retained copy that survives disk loss, a bad change, or a corrupted guest. Two recent physical-machine failures in this lab (a server NIC, and the Wazuh host losing power) made the point concretely: **anything not backed up is one hardware fault away from a rebuild.**

What was configured (2026-08-30)

- 1. **Wiped** `/dev/sdb` and created a **Directory storage** named `backup` (ext4, mounted at `/mnt/pve/backup`), content type *VZDump backup file*.

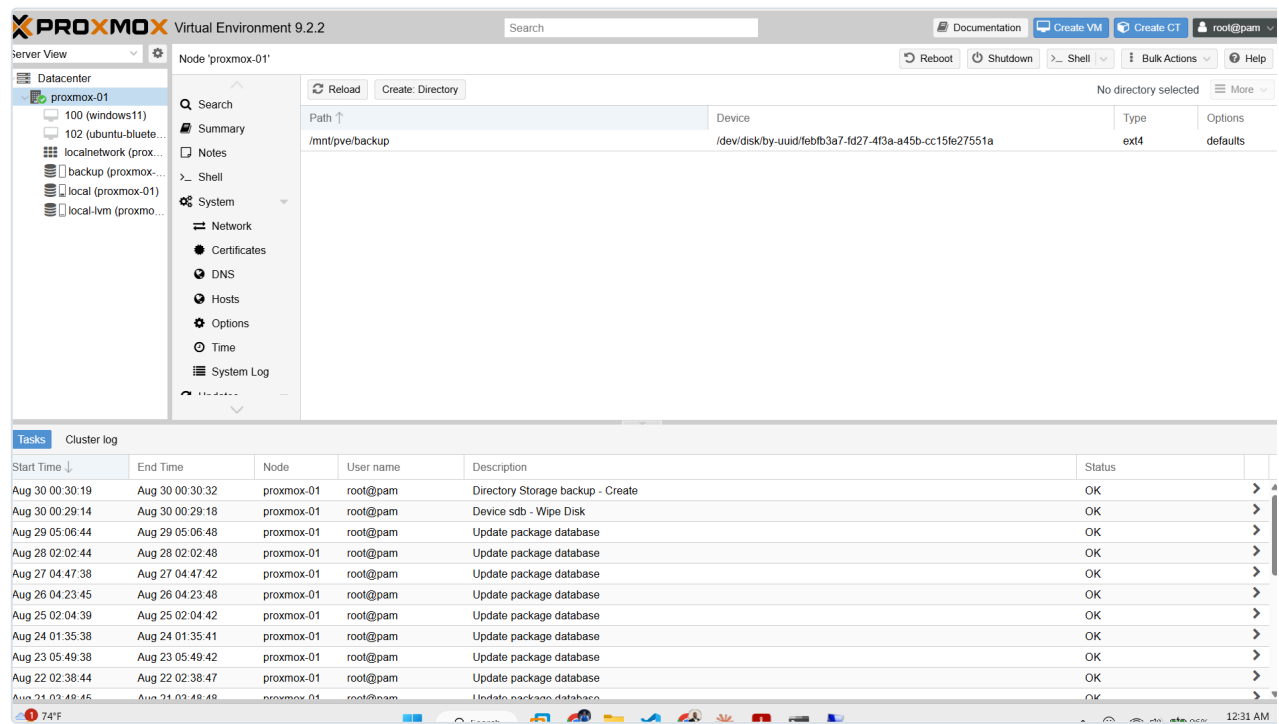


Figure 14.3: The `backup` Directory storage mounted at `/mnt/pve/backup` on the ext4 `sdb` disk. This is the dedicated backup target, separate from the disk that holds the running guests.

2. Scheduled backup job:

Setting	Value	Reason
Schedule	Daily 02:00	Off-hours, low activity
Selection	All guests	Auto-includes future VMs/CTs; no rule to forget
Mode	Snapshot	Backs up running guests with minimal disruption
Compression	ZSTD	Fast with good ratio
Retention	Keep last 3	Recent restore points without filling the disk

Setting	Value	Reason
Target	backup (separate disk)	Survives primary-disk loss

3. **Verified** with an on-demand run: task completed `TASK OK`, backup files present under `backup` → Backups for both existing guests.

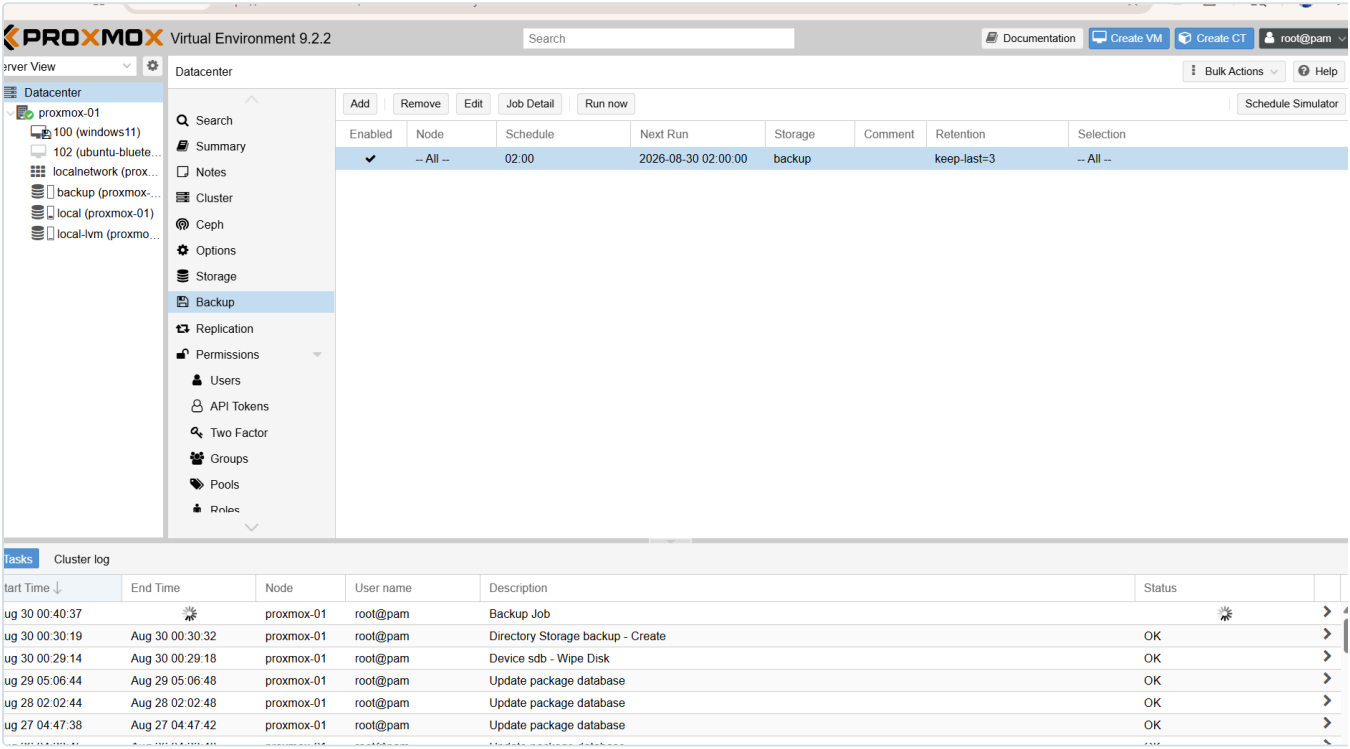


Figure 14.4: The backup job: enabled, all nodes, daily 02:00, target `backup`, retention `keep-last=3`, selection `All`. The running task at the bottom is the verification run.

What it protects, and what it does not

- **Protects:** all Proxmox VMs and containers (current and future). A lost or broken guest becomes a **restore**, not a rebuild.
- **Does not protect:** physical machines (e.g., the pfSense box, and a future physical DC02). Those need their own backup approach (for a Windows DC, the AD database is also protected by having a *second* DC, plus Windows Server Backup / system state).

Recovery

To restore a guest: Proxmox UI → `backup` storage → Backups → select the backup → Restore. Test restores periodically; a backup never verified by a restore is only a hope.

Future upgrade

Proxmox Backup Server (PBS) adds deduplication and incremental backups (far less space, faster). Not needed for a single node now; a Directory target is sufficient. Revisit if backup volume grows.

4. Capacity plan

RAM is the limit (~32 GiB). The rule that keeps the lab within budget:

Linux services run as LXC containers; Windows runs as full VMs.

LXC containers share the host kernel, a fraction of the RAM and disk of a VM. Windows cannot be a container, so it uses full VMs.

Guest	Type	Planned RAM
DC01 / DC02 (if virtual)	VM	2–4 GB
CA01	VM	2–4 GB
WKS01 / WKS02	VM	4 GB each
ANS01 (Ansible)	LXC	~1 GB
SIEM01 (Wazuh)	LXC/VM	~4 GB
MON01 (Grafana/Prometheus)	LXC	~2 GB
Kali	VM	~3 GB

All-running total is roughly 28 GB, within 32 GB, and guests can be powered on **on demand** rather than all at once. Verdict: the host comfortably carries the planned estate.

5. Machine naming convention

A consistent, role-based scheme: `<ROLE><NN>`, uppercase, no separators. The name states the role, so any machine is identifiable from its name alone without consulting an inventory. This is the basis of the asset register that NIST CM-8 (System Component Inventory) expects.

Adopted 2026-08-30 for servers. **Extended 2026-09-04** to cover the categories the original scheme left undefined: network devices, the hypervisor, and utility hosts. Those gaps were the actual source of the inconsistency, since machines with no defined category kept whatever name they were installed with.

5.1 The scheme

Category	Role prefix	Examples
Domain controllers	DC	DC01 , DC02
Certificate Authority	CA	CA01
File / SQL servers	FS , SQL	FS01 , SQL01

Category	Role prefix	Examples
Workstations	WKS	WKS01 , WKS02
SIEM / log collection	SIEM	SIEM01
Metrics and dashboards	MON	MON01
Secrets management	VAULT	VAULT01
Automation controller	ANS	ANS01
Attack / offensive	KALI	KALI01
Administrative workstation	ADM	ADM01
Privileged access workstation	PAW	PAW01
Firewall / router	FW	FW01
Switches	SW	SW01 , SW02
Hypervisor host	PVE	PVE01
Application host	APP	APP01
Utility / training host	LAB	LAB01

Two conventions worth stating explicitly, because they are the ones that get broken:

- **The number is per role, not global.** The second domain controller is DC02 , not DC10 , regardless of how many other machines exist.
- **A machine is named for what it does, not what it runs.** A Wazuh server is SIEM01 whether it runs Rocky Linux or Ubuntu. Naming after the operating system or the hardware is what produced names like nb1-core-ub01 , which tell a reader nothing about the machine's purpose.

5.2 Inventory and rename status

Current name	Target	Role	Status
DC01	DC01	Domain controller	Correct
kali01	KALI01	Attack box	Correct
nb1-core-ub01	SIEM01	Wazuh (see 5.3)	Done 2026-09-05: renamed, moved to VLAN 20 at 192.168.20.2 , cable moved to Port 4, Wazuh installed and collecting from two agents
Admin laptop	ADM01	Administrative workstation	Named at Wazuh enrolment 2026-09-06. To be superseded by PAW01 ; see the note below
lab-devops-svc01	APP01	DevOps application host	Done 2026-09-06: OS hostname and Tailscale node both renamed, Serve re-applied. See the note below

Current name	Target	Role	Status
(new Proxmox guest)	VAULT01	HashiCorp Vault	Planned, deliberately not on APP01
proxmox-01	PVE01	Hypervisor	Pending
pfSense (default)	FW01	Firewall	Pending
TL-SG108E	SW01	Managed switch	Pending
Secondary switch	SW02	Switch	Pending
TCM Ubuntu 192.168.10.4	LAB01	Training host	Pending
VM 100 windows11	WKS01	Workstation	Pending
VM 102 ubuntu-blue team	MON01	Metrics and dashboards	Pending rebuild. Grafana and Prometheus will be installed fresh rather than migrated

5.3 Role swap: monitoring hardware becomes the SIEM

The Wazuh host failed on 2026-08-29 and the monitoring host is physical while the estate's spare capacity is virtual. Rather than rebuild like for like, the two roles swap:

	Before	After
Physical nbl-core-ub01	Grafana + Prometheus, VLAN 60	SIEM01 , Wazuh, VLAN 20
Proxmox guest	none	MON01 , Grafana + Prometheus, VLAN 60

The reasoning is resource shape rather than preference. Wazuh's all-in-one deployment includes the Indexer, an OpenSearch instance whose memory floor is around 8 GB, and it holds data that must survive. Grafana and Prometheus run comfortably in about 2 GB and their state is a configuration file and a metrics database that can be restored from backup. Putting the memory-hungry, data-bearing service on dedicated hardware and the light, reproducible one on the hypervisor uses both better. The operating system stays Ubuntu 24.04 on both, which also means Wazuh moves from Rocky Linux 9.6 to Ubuntu.

5.4 Rename history

SRV1 to DC01 , 2026-08-30. Change-controlled: dcdiag /test:dns was clean before and after, the IP 192.168.50.2 and the domain ad.biira.online were unaffected, and Netlogon was restarted to re-register the DC's SRV records under the new name. The firewall alias retains its original name SRV1_DC , because it resolves an IP rather than a hostname and renaming it would invalidate the change-control history and the screenshots that reference it.

Why an ADM category exists, and why it is temporary. The administrative workstation is a daily-driver laptop that also browses the web and reads email, while holding reach into every lab VLAN (MGMT-09) and administrative access to the domain controller (MGMT-08). That makes it a Tier 0 machine by function and a Tier 2 machine by use, which inverts the trust relationship the tiering model exists to protect: the domain's

security becomes bounded by the laptop's. The proper resolution is `PAW01`, a dedicated administrative workstation with no browsing or email, with `MGMT-08` narrowed to an `ADMIN_HOSTS` alias containing only it. `ADM01` names the current reality honestly rather than pretending the laptop is something it is not. Tracked as `H-01` in `docs/13`.

Why `APP01` is not `VAULT01`. This host was originally intended to become the Vault server. It runs two web applications reachable over Tailscale, and Vault holds every secret in the environment, so co-locating them would mean a bug in an application became a path to every credential in the lab. That fails least functionality (NIST CM-7) on its face. `VAULT01` is instead built as a Proxmox guest, where it also inherits the nightly backup that a physical host does not have, which matters because Vault's encrypted store and unseal keys are the one thing that cannot be reconstructed.

Renaming a host that serves traffic over Tailscale. Three names are in play and only two of them move. The OS hostname changes with `hostnamectl`. The Tailscale machine name, which determines the MagicDNS address, is derived from the hostname and must be set explicitly with `tailscale set --hostname=` to land where you intend. The Tailscale IP does not change at all, which makes it the reliable fallback while the names settle. A `tailscale serve` configuration stores the hostname it was created with, so it does **not** follow a rename and must be reset and re-applied, after which Tailscale obtains a fresh certificate for the new name automatically. Capturing `tailscale serve status` before starting is what makes that re-application a copy of known-good values rather than a reconstruction.

A note on the hypervisor rename. `proxmox-01` to `PVE01` is the only rename here carrying real risk. Every guest configuration lives under `/etc/pve/nodes/<nodename>/qemu-server/`, so the rename involves moving a directory inside the cluster filesystem. Performed incorrectly, guests vanish from the interface. It is recoverable, but it should be done last, after a verified backup, and never at the same time as other changes.

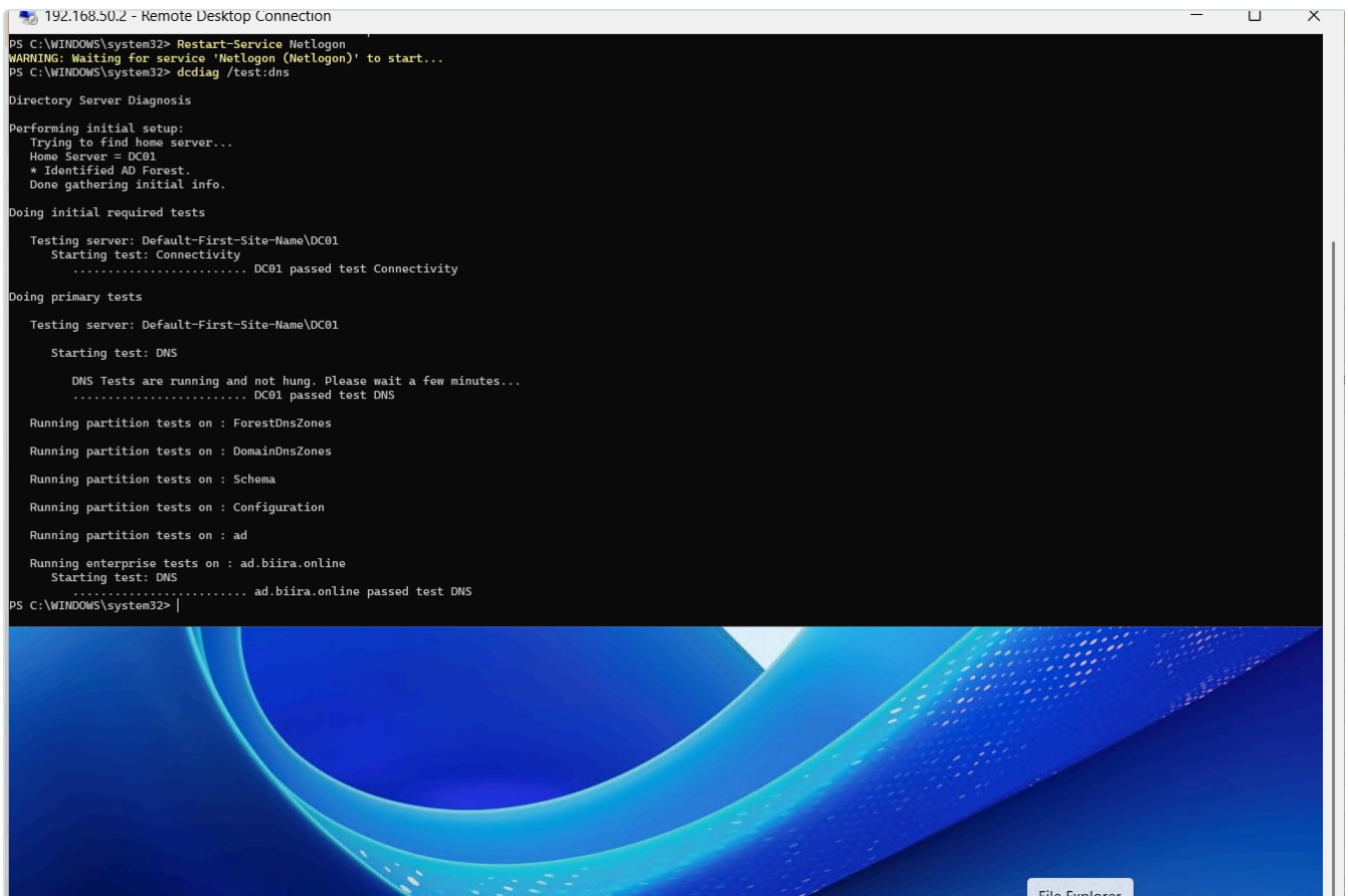


Figure 14.5: `dcdiag /test:dns` on `DC01` after the rename and Netlogon restart: Connectivity, DNS, all partition tests, and the enterprise `ad.biira.online` DNS test all pass. Evidence the rename left Active Directory healthy.

6. Standards alignment

What we did	Control
Independent, retained, scheduled backups	NIST SP 800-53 CP-9 (System Backup)
Documented restore procedure; periodic restore testing	NIST CP-10 (System Recovery) / CP-4 (Contingency testing)
Backups on separate disk from primary	Recoverability / defence against single-disk loss
Role-based naming for identifiable assets	NIST CM-8 (System Component Inventory)
This document itself	NIST CM-2/CM-6 (Baseline / documented configuration)

These are lab-appropriate implementations, not a claim of formal compliance.